

NutriScan AI

A Lightweight TinyML-Based Dew Computing Framework for Offline and Low-Latency Food Adulteration Detection

Major Academic Project Technical Report
Date: October 2026
Target Platforms: Native Android (Java) & Responsive Web (React / TypeScript)

Autonomous Edge Computing - Zero-Latency Dew Nodes - Food Authenticity Screening

1. Project Metadata & Title

Project Title: A Lightweight TinyML-Based Dew Computing Framework for Offline and Low-Latency Food Adulteration Detection.

Engineering Architecture: Autonomous Edge Dew Node coupled with Quantized MobileNetV3-Small TinyML Model.
Client Ecosystem: Native Android Application (Java, CameraX, SQLite) & Responsive Web Portal (React 18, TypeScript, Vite).
Cloud Gateway: Node.js & Express REST Backend with Firebase Security Rules.

2. Abstract & Keywords

ABSTRACT:
Food adulteration is an escalating public health crisis that causes acute poisoning, chronic organ toxicity, and severe nutritional deficiencies across developing nations. Traditional laboratory verification techniques (HPLC, GC-MS, spectrophotometry) are cost-prohibitive, lab-confined, and inaccessible in field transit markets. While Cloud AI architectures exist, they fail catastrophically in low-connectivity environments and introduce unacceptable network latency. NutriScan AI resolves this limitation through a Dew Computing Framework coupled with TinyML. Utilizing INT8 quantized MobileNetV3-Small topologies, NutriScan AI performs on-device food classification, surface necrotic damage percentage segmentation, and chemical adulterant screening in sub-50 millisecond response times without requiring an active internet connection.

KEYWORDS:
Dew Computing, TinyML, Food Adulteration Detection, Edge Machine Learning, INT8 Quantization, Android Jetpack, MobileNetV3, Offline Autonomy.

3. Introduction & Problem Statement

INTRODUCTION:
The global food supply chain requires rapid, accessible verification at the point of purchase. Field inspectors and consumers in rural and transit markets lack portable, objective testing instruments.

PROBLEM STATEMENT:
Existing analytical testing relies either on centralized laboratories or cloud servers. In rural mandis, transit warehouses, and spot-check inspections, cellular network connectivity is intermittent or non-existent, causing complete failure of cloud-based AI solutions.

4. Project Objectives & Scope

PROJECT OBJECTIVES:

1. Engineer an autonomous Dew Computing node on Android smartphones executing optical food analysis offline.
2. Develop a quantized TinyML model operating under an 80ms inference budget with a storage footprint < 4MB.
3. Establish a 9-stage pipeline evaluating image validity, food category, visible damage percentage, and adulteration hazards.
4. Build a dual-client ecosystem: Native Android (Java) and Responsive Web Portal (React / TypeScript).

SCOPE OF THE PROJECT:

The framework covers 78 food commodities across 6 primary groups: Fruits, Vegetables, Grains & Cereals, Pulses & Legumes, Spices, and Other Foods (Dairy/Oils/Pantry).

5. Existing System vs. Proposed System

EXISTING SYSTEM:
Traditional verification relies on wet chemical tests (e.g., FSSAI DART) or laboratory chromatographic assays. While Cloud AI applications exist, they require high bandwidth, suffer from latency > 480ms, and risk user privacy.

PROPOSED SYSTEM:
NutriScan AI introduces a Dew Computing framework where the smartphone acts as an autonomous local server. Neural inference executes directly on ARM NEON hardware in <50ms with zero cloud dependence. Records are cached locally in SQLite and synchronized idempotently when internet connectivity returns.

6. Literature Review & Research Gap

LITERATURE REVIEW:

- David et al. (IEEE Micro, 2021) demonstrated that TensorFlow Lite Micro enables deep learning execution on resource-constrained microcontrollers and edge processors.
- Wang (IEEE DewCom, 2016) defined the Dew Computing paradigm, demonstrating how independent edge nodes achieve offline autonomy while retaining cloud collaboration.
- Howard et al. (ICCV, 2019) introduced MobileNetV3, utilizing hardware-aware neural architecture search (NAS) for edge vision.

RESEARCH GAP:

Prior food safety apps require constant cloud connectivity and lack on-device damage percentage segmentation combined with strict probabilistic uncertainty filtering.

7. System & Android Architecture

SYSTEM ARCHITECTURE:
The architecture comprises three tiers:
1. Dew Node (Android Smartphone): Local CameraX capture, validation, preprocessing, INT8 inference, and SQLite persistence.
2. Cloud Gateway (Node.js / Express): REST API for batch synchronization, secondary verification, and regional telemetry.
3. Web Portal (React 18 / TypeScript): Laboratory visualization, interactive performance benchmarks, and consumer screening.

ANDROID ARCHITECTURE (JAVA):
Built in pure Java with Material 3 components, CameraX lifecycle management, Android Photo Picker, and DatabaseHelper SQLite persistence.

8. Website & Database Architecture

WEBSITE ARCHITECTURE (REACT & TYPESCRIPT):
Constructed with React 18, TypeScript, and Vite. Implements HTML5 Canvas pixel analysis and WebRTC video streaming for browser-based screening.

DATABASE ARCHITECTURE:
1. Local Dew Database: SQLite / Room database caching scan records offline with syncStatus (PENDING, SYNCHRONIZED, FAILED).
2. Cloud Firestore: Structured under /users/{userId}/scans/{scanId} with strict user-level access rules forbidding cross-user data leakage.

9. Food-Classification & Visible Condition Workflow

FOOD-CLASSIFICATION WORKFLOW:
Stage 1: Image Validation (sharpness, exposure)
Stage 2: Food Object Detection
Stage 3: Category Identification across 78 registered commodities.

VISIBLE CONDITION & DAMAGE PERCENTAGE ASSESSMENT:
The engine evaluates surface pixel deviations from healthy peel/flesh to calculate exact Damage Percentage:
- 0% - 6% Damage: Safe to Consume
- 7% - 25% Damage: Trim Damaged Portion Before Eating
- 26% - 55% Damage: Not Recommended for Consumption
- >55% Damage: Unsafe / Severe Spoilage

10. Adulteration Screening & Limitations

SUPPORTED ADULTERATION SCREENING:
Targets optical and colorimetric adulterant markers:
- Turmeric: Metanil Yellow azo dye, Lead chromate, Chalk powder
- Chilli Powder: Sudan Red dye, Brick powder, Rhodamine B
- Milk: Detergent, Urea, Starch, Water dilution
- Honey: Corn syrup (HFCS), Invert sugar
- Cooking Oil: Argemone mexicana oil, Mineral oil

SCIENTIFIC LIMITATIONS:
Visual screening detects chromatic shifts and texture anomalies. Colorless, odorless chemical toxins without optical manifestations cannot be detected from RGB photographs alone and require laboratory chemical assay.

11. TinyML Implementation & Quantization

TINYML ARCHITECTURE:
Backbone: MobileNetV3-Small (alpha=0.75, input resolution 224x224x3).

POST-TRAINING QUANTIZATION (INT8):
Full integer quantization mapping weights and activations from FP32 to 8-bit integers using representative calibration datasets. Compresses the model binary from 12.80 MB to 3.24 MB (74.7% reduction) while preserving 93.20% classification accuracy on hold-out test sets.

12. Dew Computing Workflow & Offline Processing

OFFLINE PROCESSING SEQUENCE:

1. User captures frame via CameraX or selects from gallery.
2. Stage 1 validates sharpness and exposure.
3. Sub-sampled decoding preprocesses image to 224x224 RGB without OOM.
4. On-device INT8 TFLite model executes in <50ms.
5. Damage percentage and edibility status computed.
6. Record committed to SQLite with syncStatus = "PENDING".
7. UI renders result cards immediately without awaiting network response.

13. Cloud Synchronization & Security

DEW-TO-CLOUD SYNCHRONIZATION:
When network connectivity is detected, background worker executes an idempotent batch POST to /api/sync/dew. Server acknowledges receipt and client marks records as SYNCHRONIZED.

SECURITY & PRIVACY:
- Zero service-account credentials embedded in client code.
- User records isolated in Firestore via request.auth.uid == userId.
- Opt-in privacy: images are never uploaded without explicit consent.

14. Mathematical Formulations & Evaluation Metrics

EVALUATION METRICS:

1. Classification Accuracy: Acc = (TP + TN) / (TP + TN + FP + FN)
2. Precision: P = TP / (TP + FP)
3. Recall: R = TP / (TP + FN)
4. Macro F1-Score: F1 = 2 * (P * R) / (P + R)
5. Laplacian Variance for Blur Detection: sigma^2 = (1/N) * sum((L(x,y) - mu)^2)
6. Damage Percentage: Damage % = (Damaged Necrotic Pixels / Total Food Pixels) * 100

15. Experimental Results & Performance Evaluation

CONTROLLED EXPERIMENTAL RESULTS:
- TinyML Dew Node (ARM NEON): Accuracy = 93.20%, Macro F1 = 0.9285, Median Latency = 38.0ms, Model Size = 3.24MB, Offline Autonomy = 100%.
- Edge FP32 Baseline: Accuracy = 94.10%, Macro F1 = 0.9370, Median Latency = 118.6ms, Model Size = 12.80MB.
- Cloud REST Baseline: Accuracy = 95.80%, Macro F1 = 0.9540, Median Latency = 482.0ms, Offline Autonomy = 0% (Fails without internet).

The Dew Computing framework reduces latency by 92.1% while maintaining zero server dependence.

16. Installation Instructions & Operating Guide

WEB APPLICATION:
cd "Major Project NutriScan AI/web-app"
npm install
npm run dev # Starts live on http://localhost:5173/

ANDROID APPLICATION:
cd "Major Project NutriScan AI/android-app"
./gradlew assembleDebug
adb install app/build/outputs/apk/debug/app-debug.apk

ML PIPELINE:
pip install -r ml-model/requirements.txt
python ml-model/src/train.py
python ml-model/src/quantize.py

17. IEEE References & Academic Citations

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[4] Food Safety and Standards Authority of India (FSSAI), "Detect Adulteration with Rapid Test (DART)," Ministry of Health and Family Welfare, Government of India, 2021.

[5] S. Ray, "Dew Computing: The Emerging Paradigm in Edge Computing and IoT Networks," IEEE Internet of Things Magazine, vol. 3, no. 1, pp. 24-29, Mar. 2020.

18. Conclusion & Future Work

CONCLUSION:
NutriScan AI validates the viability of Dew Computing and TinyML for decentralized food adulteration screening. By enabling autonomous, sub-50ms inference on consumer smartphones, the system bridges the gap between laboratory science and everyday consumer protection.

FUTURE WORK:
1. Integrating portable near-infrared (NIR) smartphone spectrometer attachments.
2. Federated learning across Dew nodes to continuously refine adulteration detection without centralizing user images.